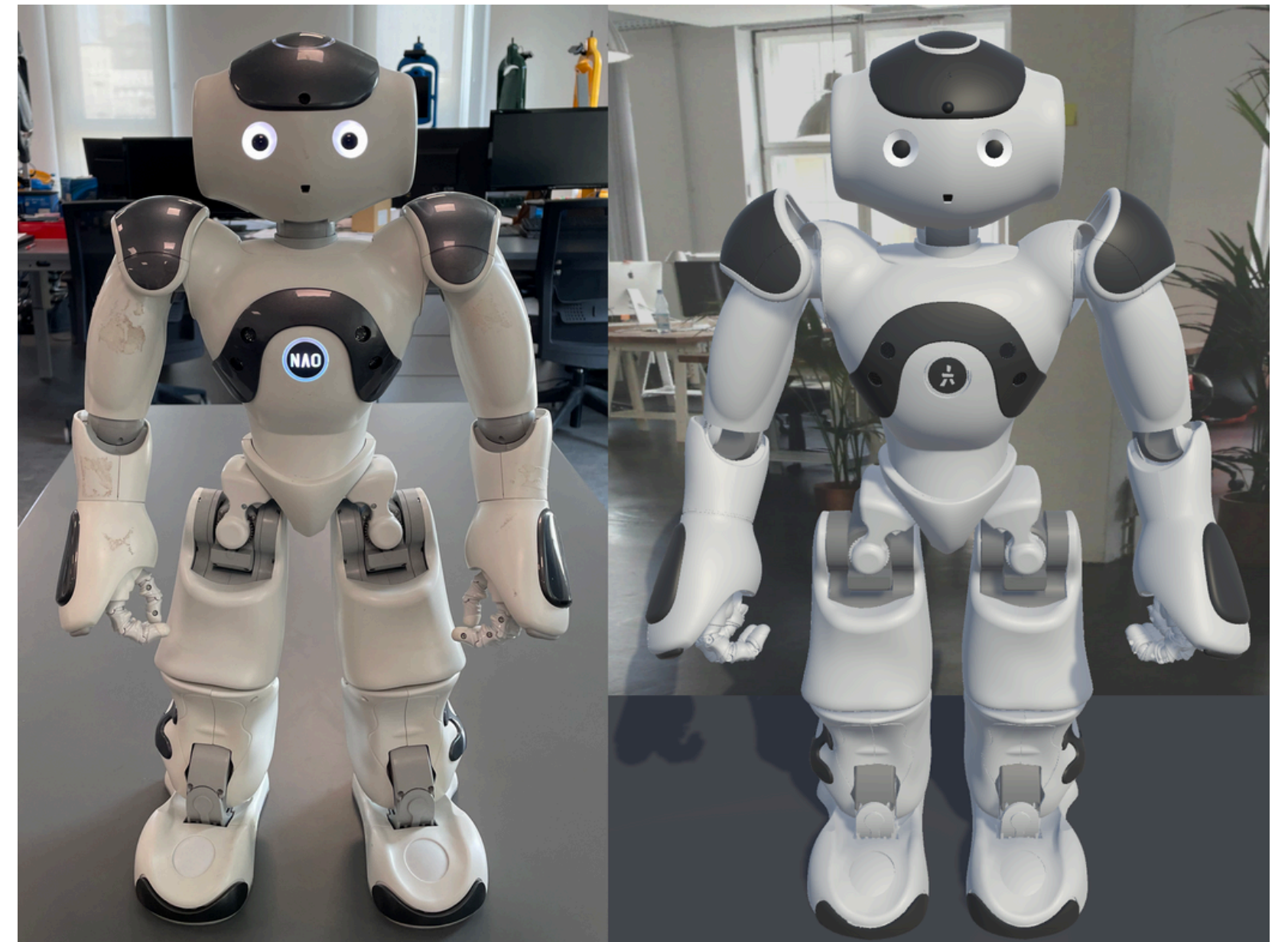


Social Presence and Emotional Interaction: Comparing the NAO Robot and Its Digital Twin

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The Challenge in Human-Robot Interaction (HRI)

- The **NAO robot** is a standard platform for HRI research, especially for studying social and emotional interaction.
- However, they are expensive and logistically complex, limiting the scale and accessibility of research.



The Unresolved Debate in HRI Research

Can a virtual robot be a valid substitute for a physical one? Existing research is conflicting.

Argument: Physical Presence is Crucial

Physical robots can elicit stronger feelings of social presence and co-presence.

- **(Li et al., 2019): Prosemics** - Humans maintain a larger personal distance from virtual robots and perceive them as more "discomforting" than physical ones.

Counter-Argument: Virtual Can Be Sufficient

For specific tasks, a well-designed virtual agent can be just as effective as a physical one.

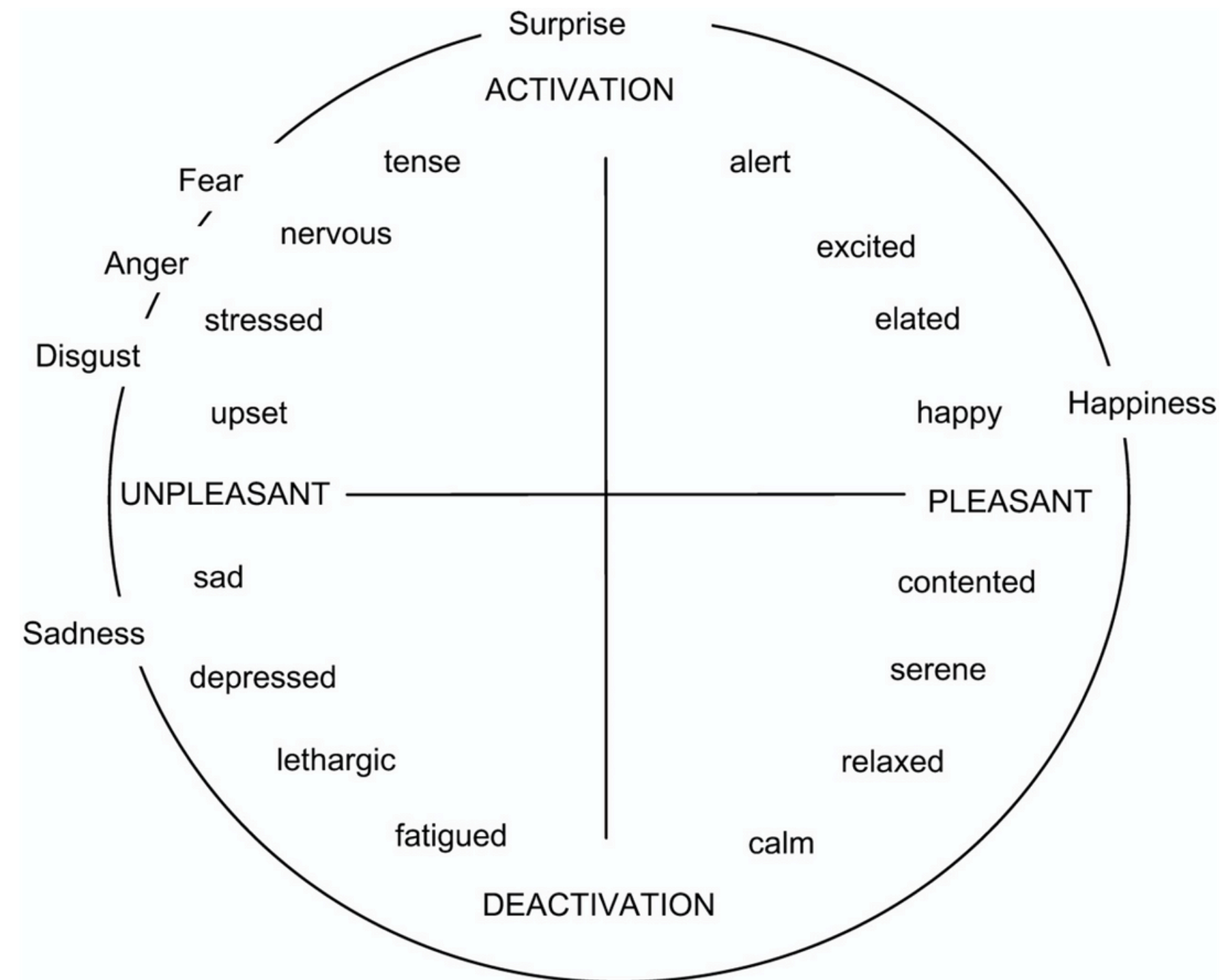
- **(Thellman et al., 2016): Trustworthiness** In a social game with NAO, perceived social presence was the key factor for user behavior, not the robot's physical form.
- **(Esterwood et al., 2025): Social presence and dialogue** - A recent meta-analysis found no significant differences in engagement or social presence between co-located and virtual robots.

Objectives of the Study

Our primary question:

- Does physical embodiment alter how humans perceive a robot's non-verbal emotional expressions?

- In social robotics, established frameworks like the Ekman/Russell models are commonly used to design and interpret emotional expressions.



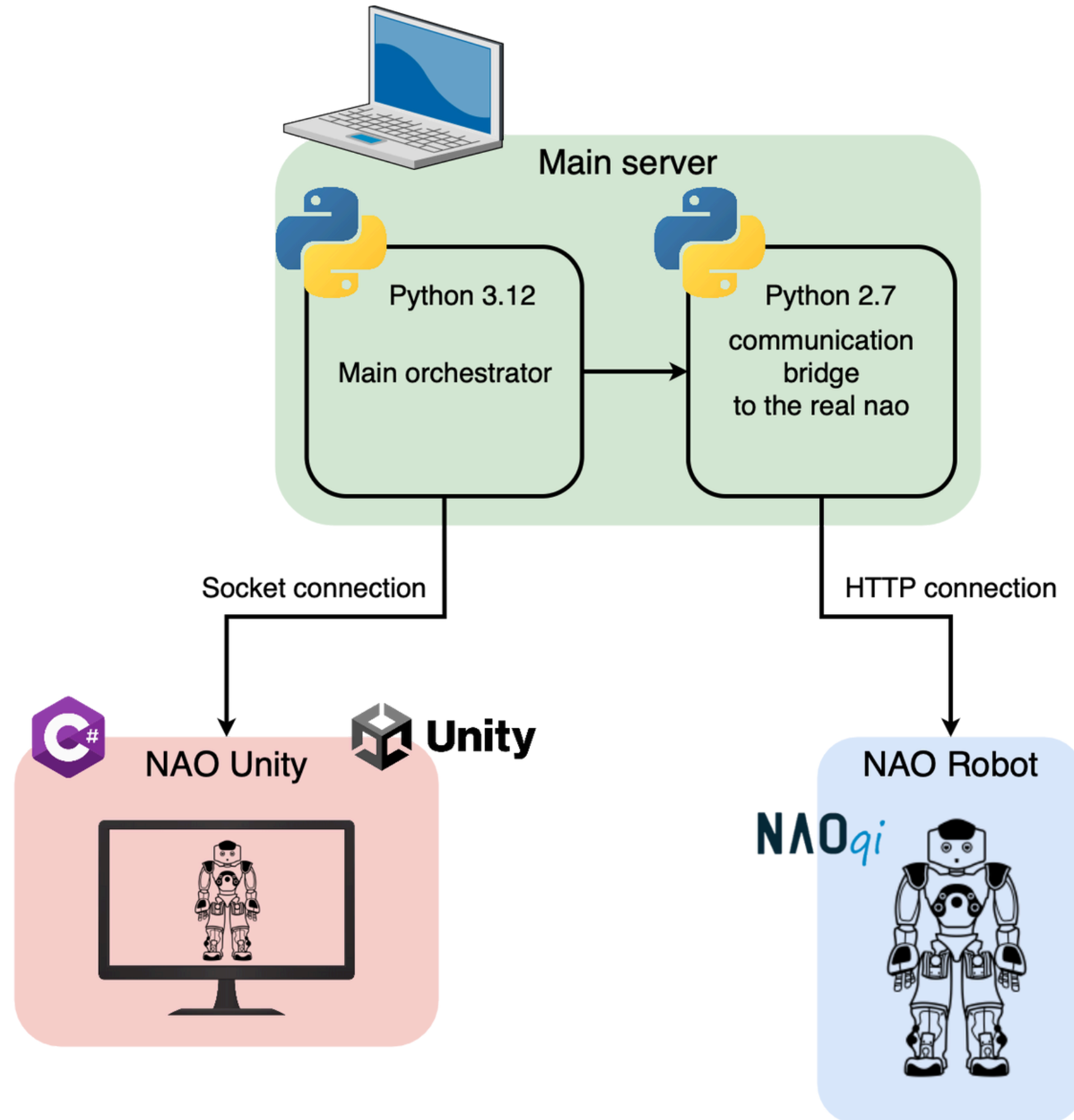
Method: The High-Fidelity Digital Twin

To isolate embodiment as the single independent variable, we have to ensure:

- **Visual Fidelity:** The virtual model is a perfect visual replica from CAD (available online) imported in Blender and Rigged.
- **Kinematic Fidelity:** Both robots share the same 25 Degrees of Freedom (DoF) and are driven by the exact same motion commands.
- Blender rigged model can be used for multiple purposes, here was imported in Unity Project



Method: Framework implementation



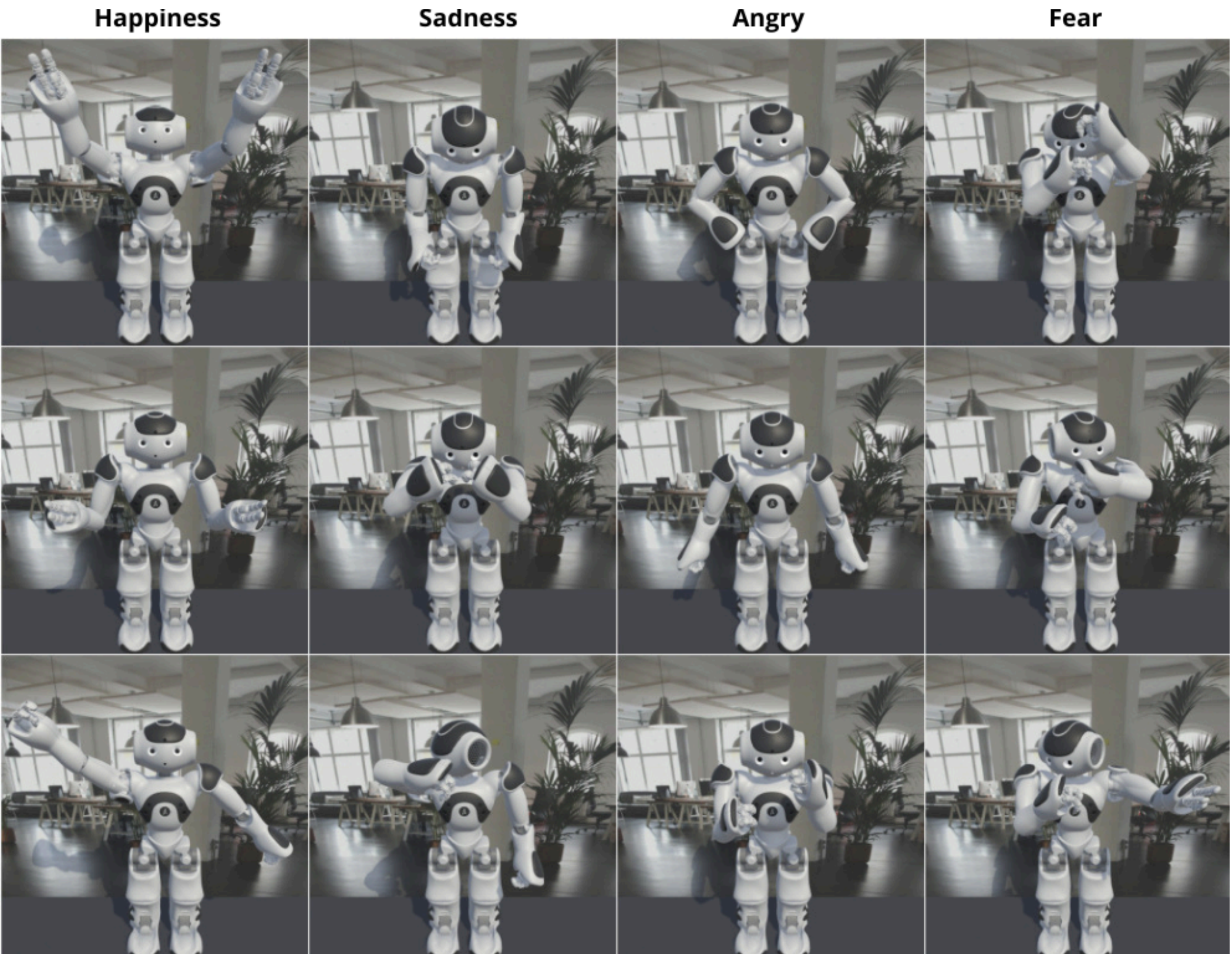
The Experimental Procedure

To isolate embodiment as the single independent variable, we have to ensure:

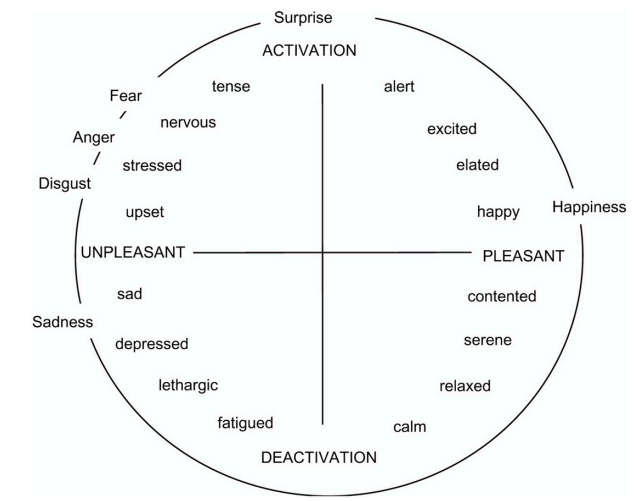
- **A set of Stimuli:** 12 distinct emotional gestures representing four basic emotions from literature: (Happiness, Sadness, Anger, Fear).

Paper	Ha1	Ha2	Ha3	Sa1	Sa2	Sa3	An1	An2	An3	Fe1	Fe2	Fe3
Häring [13]			X		X	X		-		X		
Beck [12]	X		-	X			-			X		
Cohen [16]	X			X			X					X
Erden [10]	X			X					X			
English [17]	X				X	X	X					X
Miskam [18]	X			X			X					X
Alimardani [19]	X					X	X			X	X	
Valenti [20]	X	X		X	X							
Greczek [21]	X			X			X			X		-
Consoli [22]			X	X			X			X		

- Implemented equally on real and virtual.



The Experimental Procedure



- A within-subjects, counterbalanced study to mitigate order effects.
- Participants: neurotypical adults (age 18-35) recruited from the university community.
- Procedure:
 - Each participant observed 12 gestures in total (6 on the physical robot, 6 on the virtual robot)
 - After each gesture, they identified the emotion between the four basic emotions.
 - They rated the perceived Valence and Arousal using the Self-Assessment Manikin (SAM) scale

Quale emozione riconosci?

😊 😞 😡 😠

○ ○ ○ ○

Quanto l'emozione è negativa o positiva?

1 2 3 4 5 6 7 8 9

○ ○ ○ ○ ○ ○ ○ ○ ○

Quanto è tranquillo o agitato?

1 2 3 4 5 6 7 8 9

○ ○ ○ ○ ○ ○ ○ ○ ○

ripeti azione

prossima azione

Results: Embodiment Does NOT Affect Recognition Accuracy

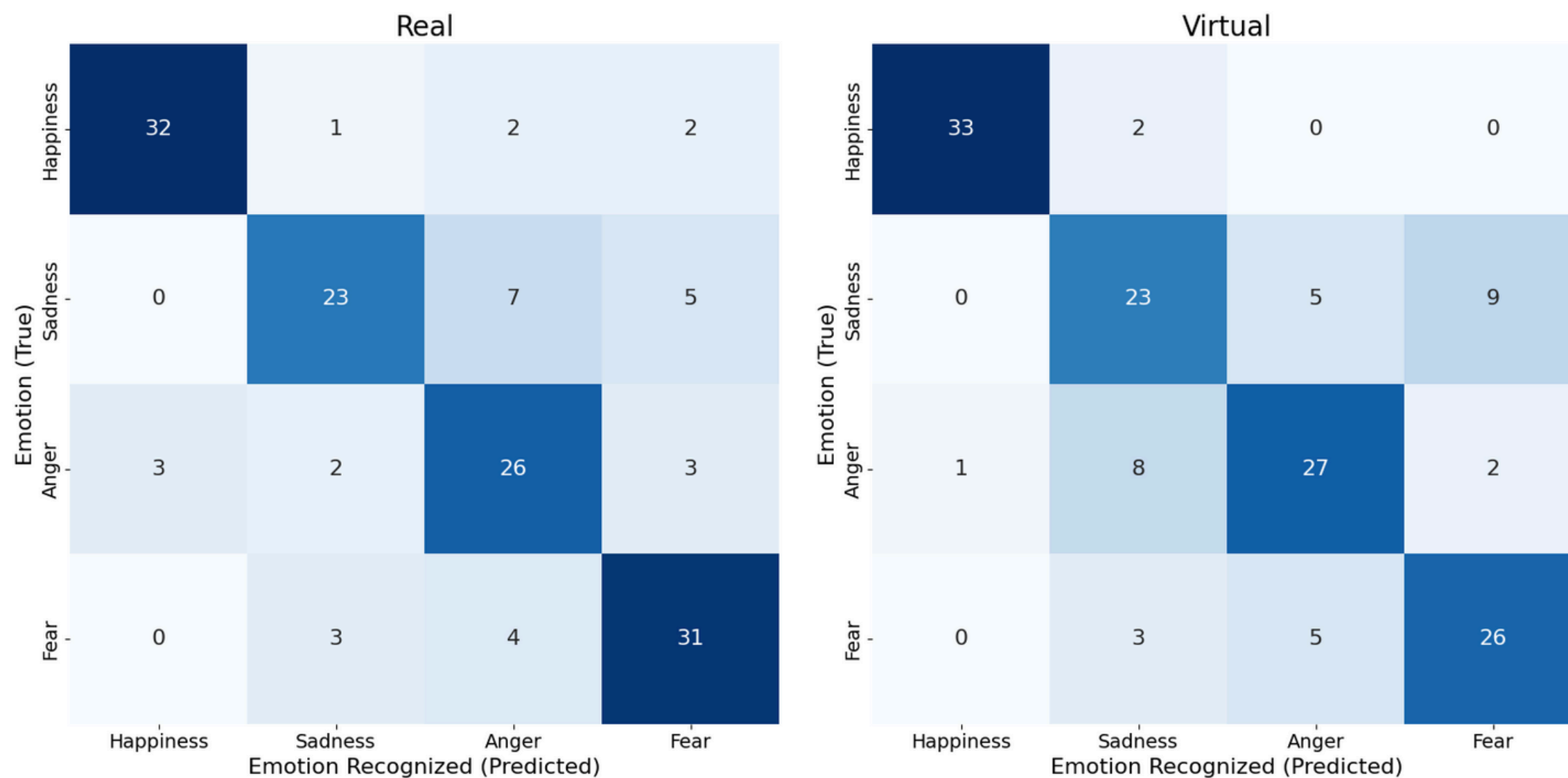
Our primary hypothesis was not supported. We found no statistically significant difference in performance.

Participants: 24 adults, 12 gestures, 288 stimuli voted

- **Physical** Robot Accuracy: **77.8%**
- **Virtual** Robot Accuracy: **75.7%**

ANOVA repeated measures confirmed the 2.1% difference is not statistically significant ($p=0.82$), far away from the minimum threshold of ($p=0.05$).

For recognizing non-verbal gestures, the high-fidelity digital twin is a **valid substitute** for the physical robot.



Stimulus	Acc. Real (%)	Errors	Acc. Virtual (%)	Errors	Acc. Combined (%)
Happiness1	100.0	–	100.0	–	100.0
Happiness3	100.0	–	100.0	–	100.0
Happiness2	61.5	angry (2), fear (2), sad (1)	81.8	sad (2)	70.8
Fear1	100.0	–	90.9	angry (1)	95.8
Fear2	76.9	angry (2), sad (1)	90.9	sad (1)	83.3
Fear3	66.7	sad (2), angry (2)	50.0	angry (4), sad (2)	58.3
Anger3	84.6	fear (2)	90.9	fear (1)	87.5
Anger1	66.7	happy (2), sad (1)	93.3	sad (1)	83.3
Anger2	75.0	happy (1), fear (1), sad (1)	25.0	sad (7), happy (1), fear (1)	50.0
Sadness1	80.0	angry (2)	78.6	fear (2), angry (1)	79.2
Sadness3	75.0	fear (2), angry (1)	58.3	fear (4), angry (1)	66.7
Sadness2	46.2	angry (4), fear (3)	45.5	fear (3), angry (3)	45.8

Results: What Actually Mattered? Gesture Legibility

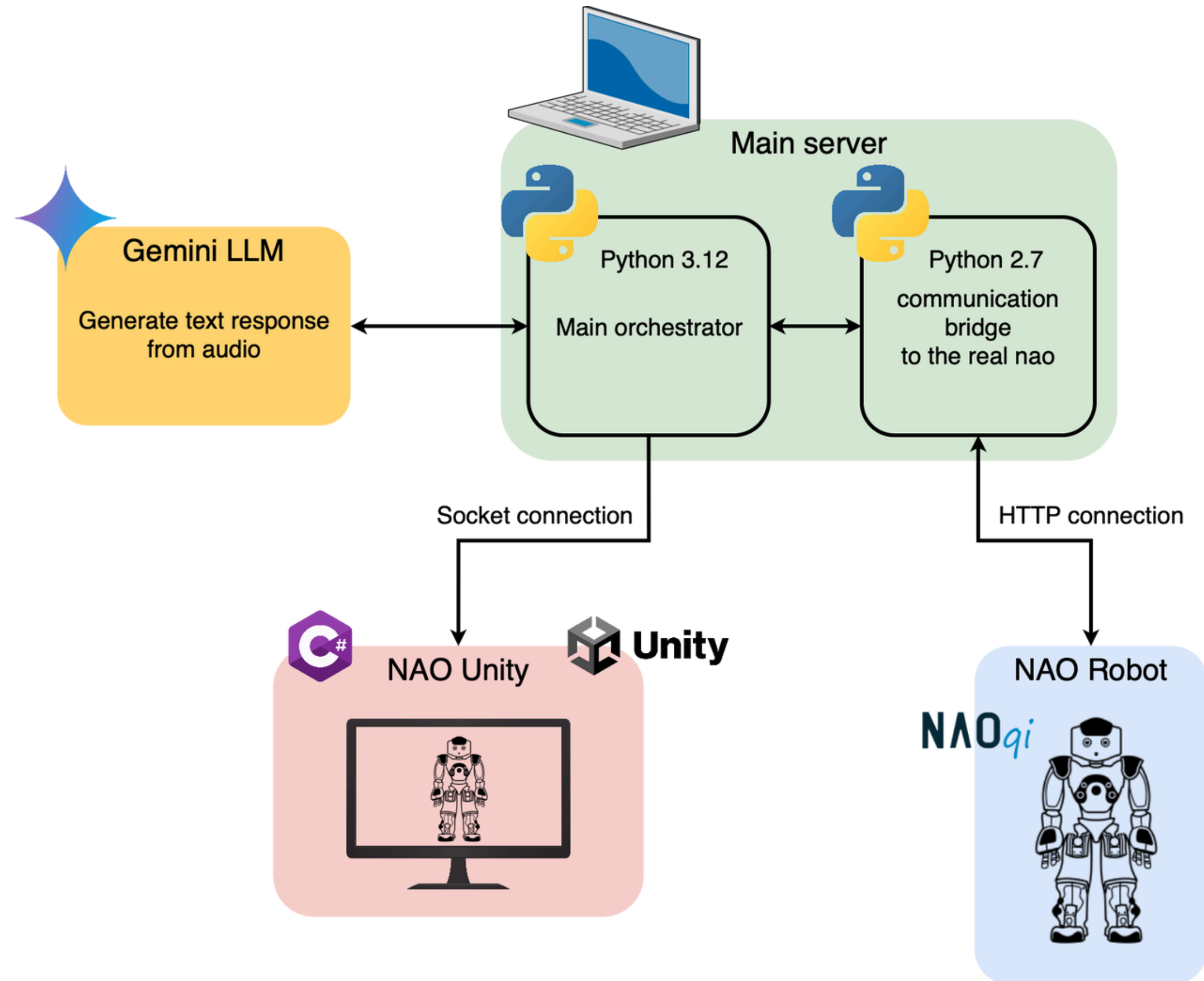
While embodiment had no effect, the intrinsic clarity of the gesture itself was the most significant factor.

- Positive Valence, Happiness was easily recognized (90.3% accuracy)
- When misclassified, negative Valence emotions were confused with each other
- Unambiguous gestures performed well on both platforms, while ambiguous gestures failed on both.

(gesture design, not physical presence, determined performance, as confirmed by ANOVA analysis - with $p=0.015$)

- However, users qualitative feedback prefer real robot, seen as more “comfortable”

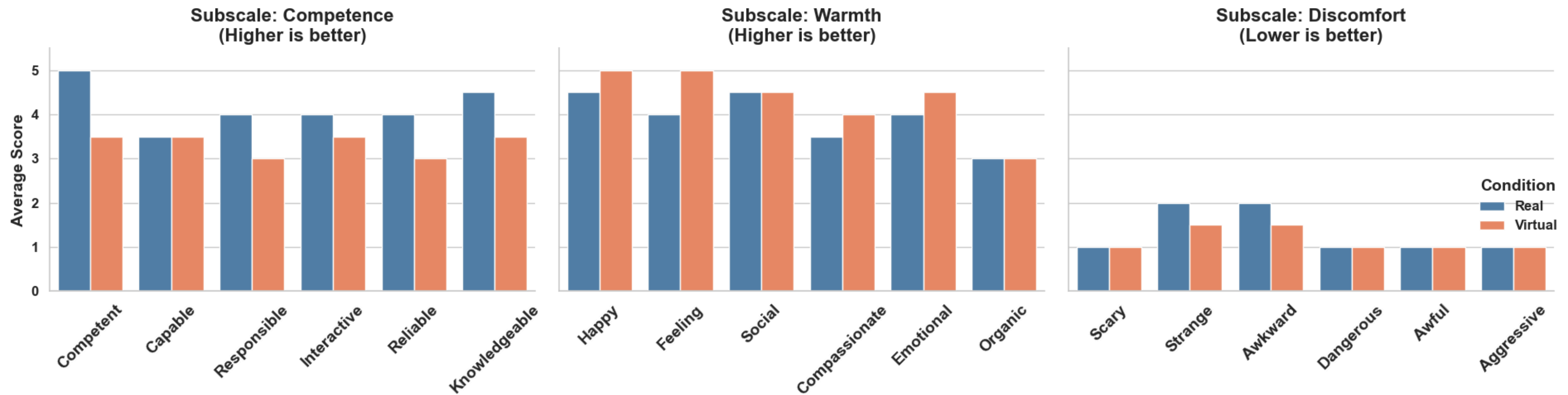
Method: Framework implementation - Extended to LLM



A first exploration: LLM-Driven Empathetic Interaction

We applied our findings in a exploratory study (N=4) using a conversational framework powered by a Large Language Model (LLM).

- Preliminary Findings (Robotic Social Attributes Scale (RoSAS) 18-item scale):
 - The **Physical Robot** was perceived as more **Competent**, despite being driven by the exact same LLM (while digital twin performed slightly better on Warmth).
 - Preliminary results - non statistically significance, more recruitment to be done



Conclusions

- A digital twin has been implemented and can be fully reused for future research.
- This study provides quantitative data to the ongoing debate: A high-fidelity digital twin is a valid substitute for a physical robot for the specific tasks such as non-verbal emotion recognition. This is a practical and cost-effective solution for HRI researchers.
- As explored in the preliminary experiment (LLM integration), this suggests a compelling direction to investigate embodiment in more complex and extended human-robot interactions (ongoing).